

Fifty-round ranking, Top-3 ensemble and reporting

P11 Independent specialist weight proposals

SYSTEM	You are one role in a blinded multi-Agent meeting selecting metric weights for AMP binary classification. Never optimize weights for a named model, a desired Top-3 or a leaderboard position.
USER	Literature, Statistics and Screening Agents independently propose all eligible metric weights from literature consensus, anonymous dataset profiles and metric-level evidence. Distinguish literature, benchmark evidence and LLM prior.
VERIFIED OUTPUT	Three role-specific proposal streams retained for the initial meeting and every round
SOURCE	agents/weight_meeting/shared_system.md · literature_agent.md · statistics_agent.md · screening_agent.md

P12 Independent methodology audit

SYSTEM	Audit unsupported evidence, hidden model preference, test-set tuning, excessive metric dominance, redundancy, ignored calibration, prevalence and unresolved disagreement. Do not produce final weights.
USER	Review all three blinded proposals and return criticisms, required changes, preferred metric directions, a leakage check and unresolved risks in strict JSON.
VERIFIED OUTPUT	reviewer_agent_audit.json plus per-round Reviewer records preserved
SOURCE	agents/weight_meeting/reviewer_agent.md · llm_agent_weight_meeting.py:457-470

P13 Chief bounded consensus

SYSTEM	Reconcile the expert proposals and Reviewer audit into the only accepted weight vector. Respond to criticisms and preserve disagreement and uncertainty.
USER	Return all exact metric keys with weights in [0.005, 0.35], sum=1, no model-specific priority and at most 0.30 L1 change from the previous accepted vector. Runtime code validates and repairs schema violations.
VERIFIED OUTPUT	chief_initial_decision.json and round_001.json through round_050.json
SOURCE	agents/weight_meeting/chief_agent.md · llm_agent_weight_meeting.py:473-497

P14 Fifty-round evidence aggregation

SYSTEM	The accepted vector scores every eligible model against the same normalized metric evidence. Model identities remain absent from weight-setting prompts, and the complete audit trail is retained.
USER	Repeat bounded evidence updates for 50 rounds, calculate each model's weighted score, then aggregate median score, IQR, median rank and Top-3 frequency without a model-specific bonus.
VERIFIED OUTPUT	50 rounds · 600 weight rows · 900 model-round scores · Top-3: pepnet_standard / amplify_imb / C_AMPs-predict
SOURCE	codex_agent_weight_meeting.py · iterative_weight_meeting.py · data/results_manual/codex_agent_weight_meeting/

P15 Exhaustive Top-3 complementarity selection

SYSTEM	This selection stage is deterministic. It must reuse stored predictions, the three dataset evaluation bundles and accepted 50-round weights without changing the metric protocol or adding model-specific bonuses. Prefer combinations supported by performance, stability and complementary errors. Do not claim leakage-free superiority while dataset independence or homology gates remain unresolved.
USER	Enumerate every unique three-model set, align samples, calculate equal-probability soft voting and score cross-dataset performance and stability using the stored evidence. Select one auditable trio from the complete combination ranking and persist the method, datasets, candidate models, metrics, ranking evidence and scientific caveat.
VERIFIED OUTPUT	816 combinations; recommended trio: C_AMPs-predict / esm-AxP-GDL / pepnet_standard
SOURCE	ensemble_top3_selector.py · data/results_manual/ensemble_top3_combination_ranking.csv · data/results_manual/ensemble_top3_selection.json

P16 Research Advisor and publication handoff

SYSTEM	Write a concise report grounded only in final weights, ranking and dataset profiles. Clearly distinguish exploratory ranking from leakage-free formal validation. Explain Top-3, weight consensus, score IQR and Top-3 frequency. Figures are a reporting layer and cannot feed values back into ranking. Every conclusion must remain traceable to JSON, CSV, Markdown, source code and run manifests.
USER	Analyze cross-dataset generalization, model limitations, dynamic metrics, next-generation AMP model directions, benchmark construction and experiment priorities. Missing evidence must be stated explicitly. Export editable vector figures, high-resolution previews, source tables, figure contracts and QA notes. Keep exploratory conclusions and external-validation requirements visible in captions and reports.
VERIFIED OUTPUT	Evidence-grounded future-directions report plus editable figures, source CSVs and QA records
SOURCE	agents/weight_meeting/research_advisor.md · agents/runtime_prompts/amp_research_advisor_template.md · figures/ · data/results_manual/ · run_manifest.py · tests/